μPIV-assisted optimization of operating conditions and device geometry for a microfluidic cavitation-microstreaming DNA extractor

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μPIV 기반 미세유체 공동 마이크로스트리밍 DNA 추출기의 구동조건 및 디바이스 형상 최적화

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Abstract

We advance a microfluidic nucleic-acid (NA) extractor by optimizing operating conditions and device geometry via μPIV (micro particle image velocimetry). Previously, we demonstrated a novel cavitation microstreaming-based DNA extraction method. However, DNA extraction performance was inferior to gold-standard commercial kits. We attributed this to under-optimized operating conditions and device geometry, as determined by qualitative flow visualization. To address this limitation, we employed top- and side-view μPIV for quantitative optimization. First, optimizing excitation voltage at 80 V_{pp} increased maximum streaming velocity to 14 mm s^{-1}, compared to 9 mm s^{-1} at the previous condition of 60 V_{pp}, boosting DNA concentration by 18.6%. Second, increasing chamber height from the previous 0.5 mm to 1.5 mm induced 3D out-of-plane flow (3OPF). Side-view analysis confirmed this enhancement, showing a velocity increase from 1.9 mm s^{-1} (0.5 mm) to 8 mm s^{-1} (1.5 mm). This geometric optimization improved DNA concentration by an additional 10.3%, achieving ~90% efficiency relative to that of a commercial kit with high purity (1.93) and successful PCR validation. We anticipate that our enhanced DNA extraction method will be widely employed for real-world NA analysis applications.

Keywords: ***μPIV, Micromixing, Microfluidic DNA extractor, Cavitation microstreaming, Optimization***

1. Introduction

Cell lysis and DNA extraction are fundamental to various applications, including infectious-disease diagnostics, gene-expression studies, virology, and bacteriology [1]. Microfluidic DNA extraction offers a rapid, portable alternative to the bench-top counterpart and often employs chemical lysis for its simplicity and excellent performance. This method requires effective mixing in multiple assay steps, posing a significant challenge in the microfluidic laminar flow regime (Re < 1) because mass transfer is limited to slow molecular diffusion. Therefore, effective microscale mixing is critical for microfluidic DNA extraction.

Previously, our group adopted bubble-induced cavitation microstreaming to enhance mass transfer [2] in a microfluidic DNA extractor. However, the extraction performance remained inferior to gold-standard kits due to sub-optimal operating conditions (i.e., excitation voltage) determined by qualitative flow visualization. Moreover, the shallow (0.5 mm) chambers confined streamlines to 2D, limiting 3D out-of-plane mass transfer.

To address these limitations, we propose enhancing mass transfer by optimizing excitation voltage to increase streaming velocity and microchamber height to promote 3D out-of-plane flow (3OPF) [3]. To quantitatively optimize these parameters, we employed in-house top-view and prism-coupled side-view μPIV, enabling quantitative characterization of flow fields and the 2D-to-3D hydrodynamic transition.

2. Experimental

2.1 Design and fabrication

The DNA extraction device consists of three layers: the top coverslip with inlet/outlet ports, the middle layer housing a mixing chamber with air pockets, and the bottom coverslip bonded with a PZT actuator (Fig. 1). To study 3OPF, the chamber height and diameter were varied while the volume is fixed (100 μL). The chip was fabricated via CO_2-laser and solvent-assisted bonding [2].

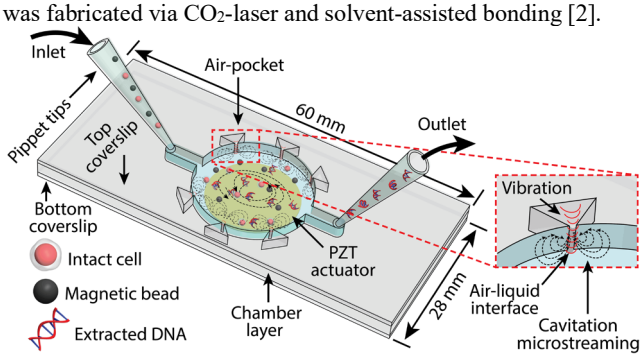

Fig. 1. Schematic of the multi-layer microfluidic DNA-extraction device employing cavitation microstreaming in a central chamber.

2.2 Top-view μPIV system

Flow velocity was quantified using an in-house μPIV system. Fluorescent tracers (6.11 μm, Spherotech) were imaged via an upright microscope (Olympus, 4× objective) and high-speed camera (Mikrotron, 281 fps). Velocity fields were computed using the PIVlab MATLAB toolbox via multi-pass FFT window deformation algorithm.

2.3 Side-view μPIV system

To quantify 3OPF fields, we developed a novel prism-coupled side-view imaging system (Fig. 2). The laser-cut chamber sidewalls were polished using sandpaper and alumina suspension to create a

clear optical interface. A right-angle prism (5 mm in width) embedded in a PDMS block was coupled to the sidewall via index-matching oil (n = 1.51), redirecting the optical path 90° to enable high-resolution vertical flow imaging (Fig. 2, red inset).

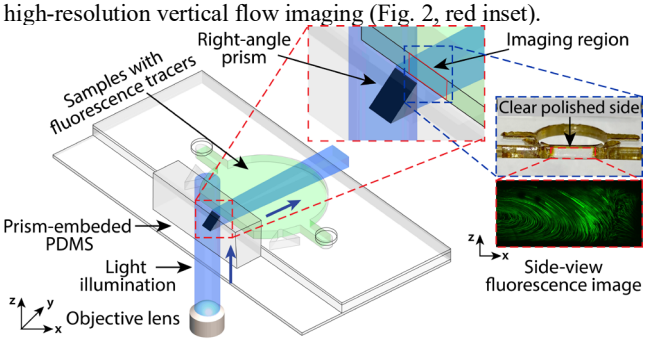

Fig. 2. Prism-coupled side-view imaging setup. Insets: polished sidewall (top) and fluorescence 3OPF streamlines (bottom).

2.4 Optimization of excitation voltage

Acoustic excitation induces cavitation microstreaming via oscillating bubbles (Fig. 1). For excitation, the resonance frequency was determined via an impedance analyzer (Zurich Instruments). Excitation voltage was optimized in a range of 20 – 80 V_{pp} to maximize streaming velocity while ensuring device stability.

2.5 Optimization of chamber design

Three chamber configurations were designed: a standard shallow design (h, D)= (0.5 mm, 17.7 mm) [2] and two thicker variants (1.0 mm, 12.5 mm) and (1.5 mm, 9.2 mm). For side-view µPIV analysis, air pockets were reduced to two, and the imaging sidewall was flattened (Fig. 2, blue inset). This isolated the flow behavior generated by a single air pocket, ensuring a clear characterization of how chamber dimensions influence the resulting flow patterns.

2.6 Cell lysis and DNA extraction

DNA extraction followed our established protocol, comprising cell lysis, magnetic collection, washing, and elution [2]. K562 human myeloid leukemia cells served as the test sample. Final DNA concentration and purity were quantified using a spectrophotometer (Implen). The DNA quality for downstream analysis was validated via PCR and gel electrophoresis.

3. Result and discussion

3.1 Optimization of excitation voltage

Increasing V_{pp} intensified bubble oscillation, boosting the maximum velocity from 1.2 mm s^{-1} (20 V_{pp}) to 14 mm s^{-1} (80 V_{pp}) (Fig. 3). The velocity for our previous under-optimized condition (60 V_{pp}) was 9 mm s^{-1}. We identified 80 V_{pp} as the optimal as exceeding this value caused bubble collapse and excessive heat.

3.2 Optimization of chamber geometry

Increasing chamber height promoted 3OPF (Fig. 4). Side-view µPIV revealed that while the 0.5-mm chamber retained planar flow, the 1.5-mm geometry generated curved streamlines confirming z-directional flow with increased flow velocity (1.9 to 8 mm s^{-1}).

3.2 Cell lysis and DNA extraction

Optimizing excitation voltage boosted DNA concentration by 18.6%. Furthermore, the 1.5-mm-thick chamber outperformed the 0.5-mm baseline case by enabling 3OPF, yielding an additional ~10% increase. The optimized device achieved ~90% efficiency relative to a commercial kit, with high purity (~1.9) and successful PCR amplification of the vWF gene (Fig. 5) from the extracted DNA. Lastly, the assay was completed under 25 minutes.

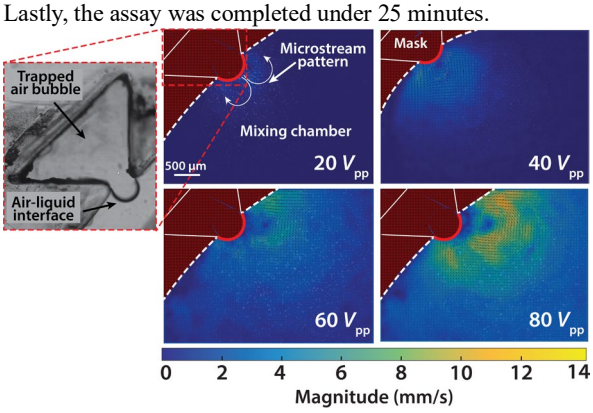

Fig. 3. Optimization of excitation voltage in a range of 20 – 80 V_{pp} using µPIV velocity magnitude fields.

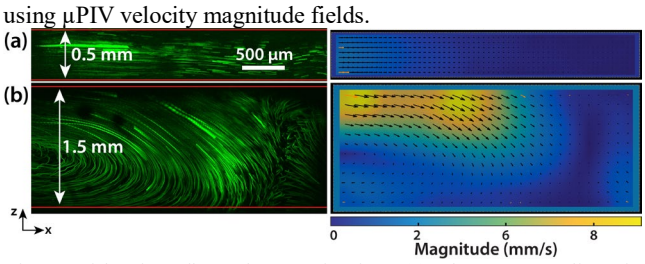

Fig. 4. Side-view flow characterization. (a) Planar streamlines in the 0.5-mm-thick chamber. (b) Recirculating streamlines in the 1.5-mm-thick chamber, indicating the emergence of 3OPF.

(a)

Method	Mixing chamber thickness (mm)	Excitattion voltage (V_{pp})	Concentration (ng/µL)	Purity (A_{260}/A_{280})
Microfluidic chip	0.5	60	8.6 ± 0.22	1.83 ± 0.04
		80	9.7 ± 0.3	1.84 ± 0.05
	1.0	80	10.5 ± 0.31	1.94 ± 0.05
	1.5	80	11.25± 0.18	1.93 ± 0.08
Commercial kit	40*	Vortex mixer	12.54 ± 0.22	2.00 ± 0.07

* Microcentrifuge tube

(b)

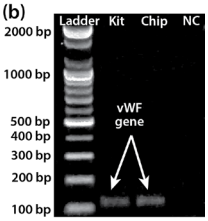

Fig. 5. DNA extraction performance. (a) Concentration and purity data for microfluidic chips vs. commercial kit (n = 5) (b) PCR and gel electrophoresis confirming excellent DNA quality.

4. Conclusion

We substantially improved cavitation-microstreaming DNA extraction performance through µPIV-assisted optimization of excitation conditions and device design. Increasing chamber height (0.5 to 1.5 mm) induced a 2D-to-3D hydrodynamic transition, thereby enhancing mass transfer for the DNA extraction. We are currently investigating this transition using a mathematical model. With excellent extraction efficiency and rapid assay time, the optimized device offers a robust solution for nucleic acid analysis.

Acknowledgments

This work is supported by the Ministry of Science and ICT of Korea (2022R11A2C1002729) and the Ministry of SMEs and Startups of Korea (RS-2024-00506893).

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